

**Experimental Manipulation of Time Pressure in a Random Dot Motion Task:
Effects on Reaction Time and Accuracy**

Group 3 Section 302

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Abstract: (Written and edited by: N. Pinka)

The Speed-Accuracy Tradeoff is a foundational principle in perceptual decision making. Understanding how individuals decide to prioritize speed or accuracy can tell us more about the decision-making process. Using a modified Random Dot Motion (RDM) task, 29 participants were assigned to one of three between-subjects conditions: (1) No timer and no speed instructions (control), (2) visible timer without speed instruction (implicit pressure), and (3) visible timer with explicit speed instructions (explicit pressure). Reaction time and accuracy were measured across 10 trials of the RDM task per participant at a 5% coherence level. We hypothesized that increased pressure would produce faster but less accurate responses, with the strongest effect seen in the explicit condition, a smaller effect in the implicit condition, and no effect in the control condition. Contrary to our predictions, one-way ANOVAs revealed no significant effects of condition on reaction time, $F(2,26) = 0.42$, $p = .663$, or accuracy, $F(2,26) = 0.65$, $p = .532$, and no significant linear trends were observed. Findings suggest that explicit and implicit time pressure cues may be insufficient to alter decision behavior under perceptual uncertainty. Further research should be done with larger sample sizes.

Experimental Manipulation of Time Pressure in a Random Dot Motion Task: Effects on Reaction Time and Accuracy (Introduction written and edited by: N. Pinka)

The speed-accuracy tradeoff is a well-studied fundamental feature of human decision-making. This tradeoff demonstrates an inverse relationship between speed and accuracy, where fast decisions are often less accurate than decisions made with more time (Wickelgren, 1977). In the decision-making process, an individual can adapt their decision-making process to fit their goal, prioritizing speed or accuracy depending on the circumstances under which they make their decision. While the speed-accuracy relationship has been well established, less is known about how subtle environmental cues influence decision behavior when time pressure is not explicitly enforced, but rather implied. It is possible that the perception of pressure may not arise only from explicit instruction, but also from environmental cues that result in the perception of time-pressure by an individual (Ordóñez et al., 2015).

Perceptual decision-making tasks, such as the Random Dot Motion (RDM) task, create a controllable method for exploring the effects of the perception of pressure on decision-making. The RDM task was originally developed by Oliver Braddick in 1973 (Braddick, 1973; Braddick, 1974). In the task, participants are asked to identify the direction of coherent motion among randomly moving dots. The online version of this task allows for a precise measure of reaction time and accuracy under uncertain conditions.

Perceptual decision-making in the RDM task is commonly modeled using an accumulate-to-bound framework. In this framework, sensory evidence (such as the movement of dots) is integrated over time until it reaches a decision threshold (Schall, 2001). Thus, there are two key variables that influence decision-making behavior in this model: the drift rate (the rate at which evidence is accumulated) and the decision threshold (how much evidence a person decides is needed to respond). Faster responses can either be due to lower decision thresholds (adjusted by the individual) or a higher drift rate (meaning better stimulus quality). Reduced thresholds result in faster, but often inaccurate responses, representing the speed-accuracy tradeoff. Within this framework, time pressure is assumed to lower an individual's decision thresholds, resulting in faster but less accurate responses. Prior work has shown that explicit speed instructions can induce a shift in decision threshold; minimal research has been done on the effect of implicit cues.

Perceptual decision-making involves a network of brain areas that transform sensory input into a behavioral response. Sensory evidence is initially encoded in the middle temporal (MT) visual area of the visual cortex, which is a motion-selective region that represents moment-by-moment information about the stimulus (Di Luzio et al., 2022). While the MT encodes the moment-by-moment evidence, it is not able to integrate this information over time to support decision formation. Instead, the evidence is accumulated over time in the posterior parietal cortex (PPC). Evidence exists for this in humans through neural ramping signals in the posterior parietal cortex as captured by the centro-parietal positivity (CPP) (Fong et al., 2026). CPP is an electrophysiological marker measured by an electroencephalogram, which shows that activity in the parietal cortex scales with stimulus strength and response timing. Causal evidence in primates further supports the role of the PPC in perceptual decision-making, as neural activity in

this brain area has been found to directly influence both perceptual decisions (Zhou & Freedman, 2019). During the accumulation of evidence, the dorsolateral prefrontal cortex (dlPFC) integrates the evidence and contributes to the formation of a decision by exerting top-down control and combining sensory inputs into a categorical judgement. (Lamichhane & Dhamala, 2015). The dlPFC and other frontal brain areas provide a framework in which both explicit instructions and more subtle environmental cues can influence decision-making by determining when a decision threshold is reached. Within this framework, perceived time pressure (either explicit or implicit) may alter decision-making behavior by lowering the decision threshold.

In this study, we aimed to investigate how perceived time pressure influences performance on a modified RDM task. We manipulated the presence of a visible timer and explicit speed instructions across three conditions: (1) no timer and no speed instructions (control condition), (2) visible timer without speed instruction (implicit pressure condition), and (3) visible timer with explicit speed instructions (explicit pressure condition). Each participant completed 10 trials of the RDM task at a 5% coherence level, during which both reaction time and accuracy were measured. Through this manipulation, our goal was to isolate the effects of implicit versus explicit pressure on the decision-making process.

According to an accumulate-to-bound framework, we would expect to see a significant lowering of decision thresholds in the explicit condition, leading to lower accuracy but faster response times, consistent with a speed-prioritized decision strategy and prior findings in the literature. In the implicit condition, we expect a similar, but smaller pattern. Implicit time cues, such as our visible timer, may not be enough to reliably alter decision boundaries and will likely not produce as large a threshold shift. Because of this, we hypothesized that reaction times on the Random Dot Motion task would be fastest in the explicit condition (visible timer with speed instructions), intermediate in the implicit condition (visible timer without speed instruction), and slowest in the control condition (no timer and no speed instructions). Conversely, we expected accuracy to be highest in the control condition (no timer and no speed instructions), followed by the implicit condition (visible timer without speed instructions), and lowest in the explicit condition (visible timer with speed instructions).

Methods: (Written by: R. Mathews, N. Pinka, R. Neal; edited by: N. Pinka)

A modified Random Dot Motion (RDM) task was programmed in JavaScript and HTML and deployed via GitHub (a web-based version control and code hosting platform), allowing participants to access the task via computer. Portions of the code were generated with assistance from Google's Gemini that were reviewed and modified by the researchers. The task used a between-subjects design, in which participants were randomly assigned to one of three instructional conditions: (1) no timer visible during the task and no speed instructions given on the instructional page (control), (2) a timer visible during the task, but no speed instruction presented on the instructional page (implicit pressure), and (3) a timer visible during the task and explicit instructions to answer as quickly as possible on the instruction page and above the task (explicit pressure).

A total of 29 undergraduate students at the University of Wisconsin-Madison participated in the study (Control: $n = 10$, Implicit: $n = 10$, Explicit: $n = 9$). Due to the time constraints of this project, we were unable to reach a larger sample size. The sample

size reflects the constraints of a course-based research project. Each participant completed 10 trials of the modified RDM task under one of the three conditions (control, implicit, and explicit) Participants provided their initials before completing the task to ensure no participant completed the task twice. Participants were instructed to complete the task only once; no duplicate submissions were recorded. Reaction time, accuracy, subject initials, and study condition data were collected and stored automatically in a Google Sheet.

The modified RDM task consisted of a black square that took up approximately 25% of the screen, within which white dots moved randomly in different directions. A coherence level of 5% was held across all conditions, such that 5% of the dots moved coherently in the target direction (either left or right), while the remaining 95% of dots moved randomly within the square display. At the start of the study, participants were instructed to provide their initials before proceeding to the next screen, where they were randomly assigned to a condition (control, implicit, or explicit). Then, participants viewed an instructional screen describing the task and their responsibilities. Participants were instructed to determine the direction of the coherent dots and to respond using the left or right arrow keys on their keyboard. Participants assigned to the explicit condition received an additional instruction to respond as quickly as possible.

Once participants selected the “Start Task” button, the first trial began. Each trial presented the dot motion stimulus at a 5% coherence level. For those in the explicit and implicit conditions, a timer was also displayed. Participants indicated their decision of the direction of the coherent dot motion using the left or right keys on their keyboard. Once a selection was made, participants immediately advanced to the next trial. A total of 10 trials were collected for each participant. There were no time cut-offs required for the study as all participants completed the task in under 5 minutes.

We selected a 5% coherence level to balance task difficulty and keep performance near a perceptual threshold to avoid ceiling and floor effects. Performance at very low coherence levels (such as 2-5%) remain above chance, but are very close to a 50% accuracy threshold (Palmer et al., 2005). Using a 5% coherence level thus allowed us to study how decisions are made under difficult, near-threshold conditions where most variability across conditions would be most apparent. For each trial, we recorded reaction time and accuracy, as well as instructional condition (control, implicit, or explicit). Reaction time and accuracy served as our dependent variables, while instructional condition served as our independent variable.

To analyze our results, we conducted a one-way ANOVA in RStudio and an eta squared value for the mean reaction time (averaged across 10 trials per participant) and mean accuracy (proportion correct per participant). We then followed up with linear trend analyses for both reaction time and accuracy, then utilized a pairwise comparison post hoc test (including 95% confidence intervals and Cohen’s *d* value measurements) to examine and compare the three different conditions’ results. Statistical significance was set at a *p*-value of .05.

Results and Discussion: (Written by: R. Neal and N. Pinka, edited by: N.Pinka)

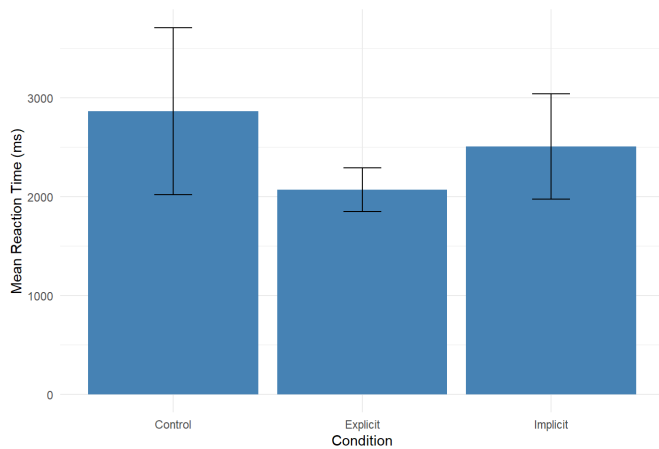
Our final sample size was $n = 29$ (Control: $n = 10$, Explicit: $n = 9$, Implicit: $n = 10$). Data were analyzed in RStudio. Data exported from the Google Sheets were imported into the coding software via .csv file, which was then read, identified, and inspected by

the software. A total of 290 trials were included across the three conditions, with a mean reaction time of 2495ms and a mean accuracy of 73%.

A one-way analysis of variance (ANOVA) was conducted, evaluating the mean reaction time and mean accuracy across groups. In both circumstances, the ANOVA showed no significant differences between groups for either reaction time ($F(2, 26) = 0.42, p = .663$) (Figure 1) or accuracy ($F(2, 26) = 0.65, p = .532$) (Figure 2). Effect sizes were estimated using eta squared (η^2), revealing small effects for both reaction time ($\eta^2 = 0.03$) and accuracy ($\eta^2 = 0.05$). The small effect sizes (3-5% of variance explained by condition), showed that our results were not statistically significant.

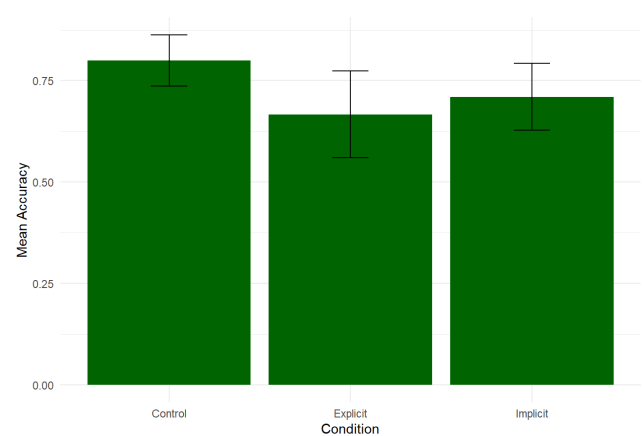
Linear trend analyses similarly showed no significant linear or quadratic trends for reaction time (condition.L: $p = .667$; condition.Q: $p = .425$) (Figure 3) or accuracy (condition.L: $p = .450$; condition.Q: $p = .409$) (Figure 4).

Figure 1
Reaction Time by Condition



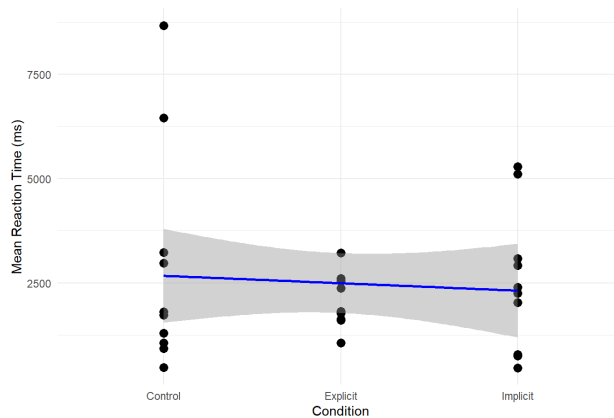
Note. Histogram of reaction time averages across Control, Explicit, and Implicit conditions. Group means were compared using a one-way ANOVA

Figure 2
Accuracy by Condition



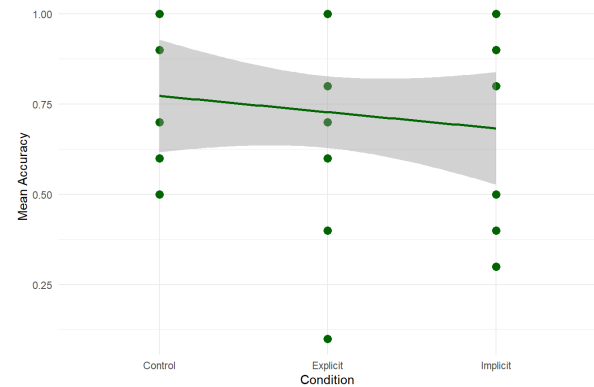
Note. Histogram of accuracy averages across Control, Explicit, and Implicit Group means were compared using a one-way ANOVA

Figure 3
Linear Trend in Reaction Time Across Conditions



Note. Scatter plot with a linear trendline for mean reaction time across Control, Explicit, and Implicit conditions.

Figure 4
Linear Trend in Accuracy Across Conditions



Note. Scatter plot with a linear trendline for mean accuracy across Control, Explicit, and Implicit conditions.

Post hoc analyses were conducted using pairwise comparisons to analyze differences between group pairs. For reaction time, all p -values were non-significant (Control - Explicit: $p = .637$; Control - Implicit: $p = .907$; Explicit - Implicit: $p = .870$), as were results for accuracy (Control - Explicit: $p = .518$; Control - Implicit: $p = .726$; Explicit - Implicit: $p = .931$). For reaction time, Cohen's d values were small (Control - Explicit: $d = 0.42$; Control - Implicit: $d = 0.19$; Explicit - Implicit: $d = -0.23$), and all confidence intervals included zero (Control - Explicit: $[-0.53, 1.37]$; Control - Implicit: $[-0.73, 1.11]$; Explicit - Implicit: $[-1.18, 0.72]$), indicating no reliable group differences. A similar pattern was observed for accuracy, with small-to-moderate effect sizes (Control - Explicit: $d = 0.51$; Control - Implicit: $d = 0.34$; Explicit - Implicit: $d = -0.17$) and confidence intervals that reflected no statistical significance (Control - Explicit: $[-0.45, 1.46]$; Control - Implicit: $[-0.58, 1.27]$; Explicit - Implicit: $[-1.11, 0.78]$).

The results did not provide sufficient evidence to support our hypotheses regarding reaction time or accuracy. Across pairwise comparisons, differences between the three conditions were small and variable. Effect sizes were consistently small-to-moderate, and all confidence intervals contained zero, suggesting no significant differences between groups. Overall, this pattern provides no evidence for the influence of instructional pressure on reaction time or accuracy in perceptual decision-making.

Within an accumulate-to-bound framework, these null results suggest that our instructional manipulation of pressure may not have been enough to alter decision thresholds. Specifically, the absence of differences in reaction time and accuracy across conditions provides no evidence of systematic threshold modulation. This contrasts with previous findings in the field demonstrating that explicit speed instruction can shift decision thresholds, as we did not observe evidence of such modulation in our study.

Several limitations may account for these findings. One of the most notable limitations is the sample size. A study with an $n = 29$ may be severely underpowered and therefore more vulnerable to influences from individual differences (e.g. competitiveness, performance anxiety, decision-making thresholds, etc.) that could have skewed the data. Secondly, participants may not have strongly engaged with the speed instructions. The sample consisted of college students and our testing window fell within two weeks of finals, a period that is commonly associated with increased cognitive load. Additionally, data were collected across a variety of environments and locations, which may have affected participants' ability to focus on the task. Finally, regarding the task itself, we acknowledge that a 5% coherence level may have been too difficult for participants, leading them to guess motion direction. If this occurred, differences between conditions would be less apparent, limiting the ability to detect instructional influences.

Future studies aiming to measure implicit time pressure on perceptual decision-making should address several methodological considerations. First, studies should gather a larger sample size and test participants in controlled environments with minimal distractors. Second, testing multiple coherence levels across conditions to examine its effect on pressure could reveal a more nuanced relationship, particularly in the implicit condition. Third, future research may consider utilizing a within-subjects design in which participants are assigned to all conditions to reduce the influence of individual differences.

In summary, this study explored the potential influence of implicit pressure on perceptual decision making tasks. Although results were not statistically significant, graphical representations of the data showed potential trends in the data. With an updated methodology, a more controlled environment, and larger sample sizes, future work may be better designed to detect effects of this nature. The potential effect of implicit time pressure on perceptual decision-making could be influential in explaining real-world phenomena, and give us a better understanding of how individuals' decisions are influenced by pressure they place on themselves. Further research is needed to clarify whether significant effects can be determined under more robust research conditions, as well as the mechanisms that may underlie decision making under implicit pressure.

Neuroscience Extensions: (Written by: R. Mathews; Initial Research by: N. Pinka)

The behavioral paradigm used in this study could be extended to include neural recordings to directly examine how perceived time pressure alters decision-making processes. Because the Random Dot Motion (RDM) task has a strong neural basis, it provides a framework for linking behavioral outcomes to underlying mechanisms of evidence accumulation and decision threshold adjustment.

One approach would be to use electroencephalography (EEG), which offers high temporal resolution (Light et al., 2011). A key signal of interest is the CPP, an event-related potential that reflects the accumulation of sensory evidence over time (O'Connell et al., 2012). Measuring CPP across conditions would allow us to determine whether perceived time pressure affects the rate of evidence accumulation or lowers the decision threshold. Additionally, functional magnetic resonance imaging (fMRI) could be used to complement this approach by identifying brain regions sensitive to time pressure. Activity in the PPC (associated with evidence accumulation) and dlPFC

(involved in top-down control and decision-making adjustments) would be of particular interest (Heekeren et al., 2004). Increased dlPFC engagement under pressure would suggest strategic adjustments in decision-making.

Under increased perceived time pressure, the decision threshold is expected to decrease. In EEG data, this would appear as reduced CPP amplitude at response, indicating that decisions are made with less accumulated sensory evidence, consistent with an accumulation-to-bound framework (O'Connell et al., 2012). The rate of CPP buildup may increase under explicit pressure, reflecting faster accumulation of evidence, or remain unchanged if the primary mechanism is threshold adjustment rather than accumulation speed (O'Connell et al., 2012). fMRI data would be expected to show greater engagement of the PPC and dlPFC under increased pressure conditions, reflecting modulation of both evidence accumulation and top-down control processes (Heekeren et al., 2004).

References: (Written by: N.Pinka; Editing by: R. Mathews)

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Contributor Block:

Programming: N. Pinka (initial code), R. Mathews (debugging - GitHub iterations + Google Sheets integration)

Presentation: N. Pinka (Course connections, experimental paradigm, and slide design), R. Mathews (hypothesis, demo, task explanation, future directions), R. Neal (statistical analysis, results, discussion, further directions)

Title: N. Pinka

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Introduction: N. Pinka

Methods: N. Pinka (writing, sources, and editing), R. Mathews, (writing), R. Neal (portion of writing)

Results & Discussion: N. Pinka (writing and editing), R. Neal (RStudio coding, writing)

Neuroscience Extensions: N. Pinka (initial research), R. Mathews (writing)

References: N. Pinka (writing), R. Mathews (editing)